



Department of Software Engineering
University of Management and Technology (UMT), Lahore
Software Requirements Specification (SRS)
FYP-1 AI powered skin, scalp personalized treatment makeup
and fashion assistance

Version 1.0

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1. Introduction

1.1 Purpose

This document explains the goal of the AI-Based Skin and Hair Scalp Analysis System. The main purpose of this phase is to build a smart system that can look at a person's skin and scalp image and find problems like acne, dandruff, or dryness using AI. It will also give **personalized treatment according to their medical history if someone have diseases like (e.g diabetic, heart issues, pregnancy, etc.) so it cannot effect other diseases**. This document helps developers, teachers, and project members understand what the system will do and how it will work.

1.2 Scope

In this first phase, the system focuses only on skin and scalp analysis. Users can upload or scan a photo of their face or hair, and the system will:

- Find skin issues like acne, dark spots, or dryness.
 - Find scalp issues like dandruff, oiliness, or hair fall.
 - Show a simple report showing the detected problems, their severity level, and treatment recommendations based on the analysis. If the severity is high, then it suggests users consult with dermatologists.
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1.3 Definitions, Acronyms, and Abbreviations

- **AI (Artificial Intelligence):** Smart computer technology that helps find skin or hair problems.
 - **Scalp Analysis:** Checking scalp health like dryness, oiliness, or dandruff.
 - **Skin Condition:** Any visible problem on the face like acne or dark spots.
 - **Recommendation:** Personalized treatment given by the system based on your results.
-

1.4 References

- **FYP Proposal:** AI-Powered Real-Time Face, Hair PERSONALIZED TREATMENT AND MAKEUP AND FASHION ASSISTANCE.
 - **Datasets:** *kaggle*
 - **Tools:** draw.io, google coolab, clickup, visual etc
 - *Research on AI for skin and hair detection*
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1.5 Overview

This document explains everything about **Skin and Scalp Analysis**, what the system does, how it analyzes images, and what output it gives.

Later phases will include more advanced features like AR hairstyle preview, doctor booking, and multi-language or voice features.

2 Overall Description

2.1 Product Perspective

It mainly focuses on checking the skin and scalp health of a person using AI. The user will upload or scan a picture of their face or hair, and the system will automatically find skin or scalp problems and suggest personalized treatment.

2.2 Product Functions

The main things this system can do are:

- Take or upload a **photo** of your face or hair.
 - Find **skin problems** like acne, dark spots, or dryness etc.
 - Find **scalp problems** like dandruff, oiliness, or hair fall etc.
 - What problems were found and how serious they are and treatments such as (*Acne*: Use gentle face wash, aloe vera gel, *Dark spots*: Use vitamin C serum, *Dry skin*: Use moisturizer and drink more water, *Dandruff*: Use anti-dandruff shampoo, *Hair fall*: Apply hair oil)
 - The system gives **AI**-based care suggestions to help improve the user's skin and hair condition. It tells users what daily care routine, products, or natural remedies they can use.
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2.3 User Characteristics

User Roles:

User – Users log in, upload images, and save/view reports. Admin – Manages users, system settings, AI models, and databases, Salons and the dermatologist.

People who want to check their skin or hair/scalp condition.

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- They can find out if they have problems like acne, dark spots, dandruff, dryness, or hair fall.
 - Skills to take or upload a photo and read the result report that have knowledge about skin disease and scalp issue
-

2.4 Constraints

- The system needs clear and good-quality images to give correct results.
 - It works properly only with internet connection.
 - The accuracy depends on how well the AI model is trained.
 - It must keep user data private and safe.
 - Works best on devices with a good cameras like smartphones or laptops.
-

2.5 Assumptions and Dependencies

- It is assumed that users will give real and clear photos.
 - The system depends on **AI** trained with skin and scalp datasets.
 - Use tools like TensorFlow and PyTorch for image analysis.
 - Needs a working internet connection to process images.
 - Future updates may depend on external APIs or **expert consultations**.
-

2.6 System Environment

- **Device:** Works on a mobile phone or web
 - **Software:** Web or mobile app both with **React**
 - **Backend:** Django rest api
 - **Internet:** Needed for scanning images and showing results.
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3 Specific Requirements

3.1 Functional Requirements (FR)

FR1: User Authentication

The system shall provide secure user registration and login using email/phone + password. Users shall be authenticated via JWT-based session handling (Django REST).

FR-2: Database Storage

The system shall store:

- User profile information (name, gender, age, medical info).

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- Authentication credentials (securely hashed).
- Uploaded images (face, scalp) to secure media storage.
- Diagnostic results and reports.

FR-3: User Image Input

The system must allow users to upload or capture high-quality images of their face or scalp through a web or mobile interface.

FR-4: Image Validation

Before analysis, the system must validate image clarity, lighting, and file size (≤ 10 MB). If the image is invalid, the system prompts the user to retake or reupload. It also confirms before uploading is it scalp or skin

FR-5: AI-Based Detection Pipeline

The system shall use the following AI modules:

- Face & Scalp Detection: Mediapipe.
- Segmentation: U-Net
- Condition Classification: EfficientNet-B4+ YOLOv8
- Severity score: XGBoost
- Recommendation Engine: LLM

These models shall be integrated through Django REST + React.js + PyTorch/TensorFlow.

FR-6: AI-Based Skin Condition Detection

The AI module must detect facial skin issues such the system shall detect:

- Acne
- Dark spots
- Hyperpigmentation
- Skin dryness

using pre-trained CNN models.

FR-7: AI-Based Scalp Condition Detection

- The system must analyze scalp images to detect Dandruff
- Dryness
- Oiliness
- Indicators of hair fall

related problems.

FR-8: Severity Classification

- Detected skin and scalp conditions must be categorized by Mild
- Moderate
- Severe.

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FR-9: Personalized Diagnosis and Care Suggestions

Based on the AI results and user-provided information (age, gender, lifestyle), the system must generate custom skincare and haircare suggestions such as product use.

FR-10: Medical-History-Aware Treatment Recommendation

The system must provide personalized treatment recommendations that consider the user's medical history to ensure safety and accuracy. For example, if a user has diabetes or allergies, pregnancy, the system must automatically filter out unsafe products or routines and suggest medically appropriate alternatives.

FR-11: Result Visualization and Report Generation

The system must produce textual report showing detected issues, severity, and recommended treatments.

FR-12: Result Storage and History Tracking

The system shall provide a dashboard where the user can:

- View past diagnoses
- Track progress over time
- Compare condition improvement historically

FR-13: User Profile Management

Users shall be able to:

- Update their personal profile
- Edit medical information
- View saved reports
- Manage uploaded images

FR-14: Feedback and Model Improvement

Users should be able to rate the accuracy of the diagnosis or recommendations, helping refine AI models over time.

FR-15: Data Security and Privacy

All stored data including medical information and uploaded images must be encrypted and accessible only to authorized users.

FR-16: Error Handling

The system must display informative error messages when encountering poor quality images, network issues, or internal processing errors.

FR-17: Safe Recommendation Enforcement

The system shall ensure that all skincare and scalp care recommendations are medically safe and avoid suggesting harmful or risky products.

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FR-18: Transparent AI Output Explanation

The system shall provide a simple explanation or reasoning (e.g., highlighted region, detected patterns) about why a condition was detected.

FR-19: User Education Module

The system shall provide educational tips related to lifestyle, environment, and habits that may affect the detected skin or scalp condition.

FR-20: Quick Response Handling

The system shall notify the user if analysis takes longer than expected and shall show a progress indicator to maintain user engagement.

FR-21: Automatic Privacy Protection

The system shall anonymize user images before they are used for AI model improvement or dataset expansion.

FR-22: Regular Model Update Support

The system shall allow periodic updates of AI models to improve accuracy based on new data and user feedback.

3.2 Non-Functional Requirements (NFR)

These describe how the system should perform and behave.

- **Performance:** Results should be produced within 15–30 seconds after image upload under standard internet speed.
- **Accuracy:** The AI model must achieve at least 75–80 % accuracy on benchmark dermatology datasets.
- **Reliability:** Consistent and repeatable outputs must be guaranteed for similar images.
- **Scalability:** The Django-based backend must handle multiple concurrent users without degradation.
- **Usability:** The interface should be minimal, intuitive, and clearly labeled (“Upload”, “Analyze”, “View Report”).
- **Security:** All user data (medical, images, reports) must be encrypted in transit and at rest using SSL/TLS and AES-256.
- **Availability:** Maintain at least 99% uptime for 24/7 accessibility.
- **Maintainability:** Codebase should be modular, documented, and version-controlled (Git) for easy model retraining and updates.

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- **Compatibility:** Must run smoothly on web browsers and mobile devices; support standard formats (JPG, PNG, JPEG).
 - **Privacy:** Follow data-protection principles require explicit consent before collecting or using medical data and anonymize when used for AI training. Users must give consent before uploading personal data. Access control for Registered User, Admin. Encrypted storage and secure login (passwords, optional 2FA).
- **Portability:** Deployable on AWS or Firebase cloud environments with minimal reconfiguration.
 - **Error Tolerance:** Handle failed uploads, poor connectivity, and invalid images gracefully, prompting user correction.
 - **Health Safety Compliance:** All treatment suggestions must be medically safe, context-aware, and compliant with health-data handling standards.
 - **High Accuracy Expectation:** The system should maintain high prediction accuracy to avoid misleading or incorrect diagnoses.
 - **User Interface Simplicity:** The system interface must remain simple, and easy to use for non-technical users.
 - **Transparency Requirement:** AI decisions must be presented in an understandable manner, increasing user trust in the diagnosis.
 - **Strong Privacy Assurance:** User-uploaded images and medical data must be always protected with strict privacy measures.
 - **Performance Speed Requirement:** The system should produce diagnostic results quickly to maintain user satisfaction and usability.
 - **Continuous Improvement Expectation:** The system should support future model enhancements without requiring major system redesign.

3.3 Interface Requirements

These define how users and systems will interact.

- **User Interface:**
 - A clean and friendly screen for uploading or capturing images.
 - Buttons for “Upload Image”, “Analyze”, and “View Report”.
- **Software Interface:**
 - Use **AI libraries** for image processing.
 - Connect with a **database** (like SQL) to save user data.
- **Hardware Interface:**
 - Works on devices with a **camera** (mobile

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- **Communication Interface:**
 - Requires a **stable internet connection** for sending and receiving data.
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3.4 System Features

The system includes these main feature modules:

- **Image Upload Module:** For taking or uploading photos.
 - **AI Analysis Module:** For detecting skin and scalp issues.
 - **Report Generation Module:** For detected issues and care advice.
 - **User Result Module:** For saving or viewing past analysis reports.
-

3.5 Data Requirements

Defines how data will be handled and stored.

- **Input Data:** Image file (JPG, PNG, or JPEG format).
 - **Output Data:** Report showing skin or scalp conditions, severity, and care tips.
 - **Data Validation:**
 - The system should check if the image is clear and under 10MB.
 - If not, it should ask the user to upload a better photo.
 - **Data Storage:**
 - User images and reports should be saved securely in the database.
 - Only authorized users can view their data.
-

5 External Interface Requirements

5.1 User Interfaces

- Simple and easy design for all users.
 - Users can take or upload a photo of their face or hair.
 - Systems show a report on problems, seriousness, and treatments.
 - Buttons like Upload, Analyze, and View Report for easy use.
 - Results shown with text and color marks (e.g., red = serious issue).
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5.2 Hardware Interfaces

- Needs a camera to take clear skin or hair photos.

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- Works on mobile phones, laptops, or PCs.
 - Needs an internet connection for image analysis.
-

5.3 Software Interfaces

- **Frontend:** React.js
 - **Backend:** Django rest api
 - **AI Tools:** google coolab, clickup, visual
 - **Database:** mySQL
 - **APIs:** Used for photo upload, analysis, and user login.
-

5.4 Communication Interfaces

- The system uses the internet to send and receive data.
 - When a user uploads a photo, it is sent to the Django server through the REST API.
 - The Django server passes the photo to the AI model for checking.
 - The AI model analyzes the photo and creates a report showing skin or scalp problems.
 - The report is sent back to the Django server, which then stores it in the database and shows the result to the user.
-

6 Design Constraints

- The system is built using the following **technologies**:
 - **Frontend:** React
 - **Backend:** Django to process images and manage data.
 - **AI Tools:** Google coolab, clickup, visual
 - **Database:** MySQL to save user photos, reports, and login data.
- The system uses a **client-server structure**:

Client side (user side): The user can take or upload a photo of their face or hair and see the results.

Server side (Django): It receives the photo through the REST API, sends it to the AI model to check skin or scalp problems, then sends the report back to the user.

- Requires a camera and a stable internet connection to upload and analyze photos correctly.
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Privacy and Security:

User Consent: Users must give consent before uploading personal information, photos, or health-related data.

- Consent will be recorded, and users can withdraw it anytime.

Data Storage: All user data (personal details, skin/hair analysis, photos) will be stored securely in encrypted databases.

- Sensitive data like passwords will use strong encryption (AWS-256).

Data Protection: Only authorized users (admins or the specific user) can access personal data.

- Secure authentication mechanisms (passwords, optional 2FA) will be used.
- Regular backups and system audits ensure data integrity.

7 Non-Functional Requirements (Detailed)

7.1 Performance

- The system shall analyze hair condition (dryness, dandruff, hair loss) and scalp health within 3–5 seconds per image.
- Real-time camera-based hair/scalp scanning should run at minimum 20 FPS for smooth detection.
- **Throughput:** The system must support up to 50 hair/scalp analyses per minute on the server without degradation.

7.2 Scalability

- The module must support **up to 1000 concurrent users** performing hair or scalp scans.
- Server resources shall auto-scale based on spikes in analysis requests.
- Cloud storage (for photo history) must scale dynamically as users store hair/scalp progress pictures.

7.3 Security

- All user images (hair and scalp photos) must be encrypted end-to-end during upload and storage.

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- User medical history related to hair/scalp (dandruff history, allergies, infections) must be securely stored using role-based access control.
 - Only authenticated users can access their stored hair/scalp reports.
-

7.4 Usability

- Hair and scalp scanning UI should provide clear instructions, progress indicators, and real-time feedback.
 - AR-based hair preview must remain responsive with less than 200 ms latency.
 - The module must be accessible (large icons, simple layout, multilingual support such as English/Urdu).
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7.5 Reliability

- Hair/scalp diagnosis features must maintain 99% uptime.
 - If real-time scanning fails due to weak lighting or camera issues, the system should auto-switch to photo upload mode.
 - All scanned data such as hair condition history shall have automatic cloud backup and recovery capability.
-

7.6 Maintainability

- The hair/scalp analysis module must be built in modular components (image processing, AR preview, treatment suggestions) to allow easy updates.
 - ML models (hair loss detection, dandruff classifier) should be upgradable without affecting the main app flow.
 - Code should follow a clean architecture to simplify patching, bug fixes, and feature additions.
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8 System Evolution / Future Enhancements

In future versions, the **Skin and Hair/Scalp Analysis System** can be improved with the following enhancements:

- **Advanced Skin Condition Detection:** Include more skin diseases such as rosacea (redness, bumps, and irritation on the face) and fungal infections.
- **AI-Based Personalized Treatment Plans**

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- Generate customized skincare routines based on daily progress and environmental factors (weather, humidity, pollution).
- Recommend dermatologist-approved products automatically.
- **Real-Time Scalp Health Monitoring**
 - Add smart scalp scanning to detect early signs of dandruff, hair thinning, or infections.
 - Suggest specific shampoos, oils, or diets to improve scalp health.
- **Hair Growth and Damage Tracking**
 - Use AI to compare before/after photos to measure hair growth or recovery from damage.
 - Provide progress graphs and personalized tips.
- **Integration with Smart Devices**
 - Connect with smart mirrors or wearables to monitor skin moisture, UV exposure, and scalp temperature for better analysis.
- **Predictive Analysis**
 - Use AI to forecast future skin or hair issues based on current health and lifestyle patterns.
- **Augmented Reality (AR) Enhancements**
 - Improve live AR previews for hair color, styles, and skin tone correction.
 - Offer 3D scalp and skin visualization for educational and diagnostic use.
- **Expert Collaboration and Feedback Loop**
 - Allow dermatologists to review AI results and provide professional feedback to enhance model accuracy

9 Appendices

9.1 Appendix A: Glossary

- **Skin Analysis:** Checking the skin for acne, spots, dryness, or redness.
- **Scalp Analysis:** Studying the head skin to find dandruff, oil, or hair loss.
- **Hair Condition:** The health of hair, like smooth, dry, or damaged.

9.2 Appendix B: Data Samples

Input:

- User photo of face or scalp.
- User details: age, gender, hair type, skin type.

Output:

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- Detected skin or scalp issues (e.g., acne, dandruff).
- Treatment or care suggestions (e.g., shampoo, moisturizer).
- Health report summary.

9.3 Appendix C: Revision History